

CIFAR-10 Pre-Trained vs. ImageNet on CIFAR-10.1 v6

Group ID-1

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01 Challenge

02 Approach

03 Augmentations with Albumentations

07 Results

Small dataset (2000 images)

Distribution shifts (varied lighting, backgrounds)

High inter-class similarity

Evaluate transfer learning with the following models:

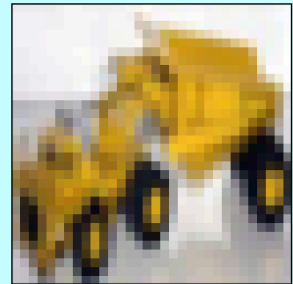
RepVGG-A2: 26.8M parameters, CIFAR-10 pre-trained, 32x32.

EfficientNetV2-S: 20M parameters, ImageNet, 384x384.

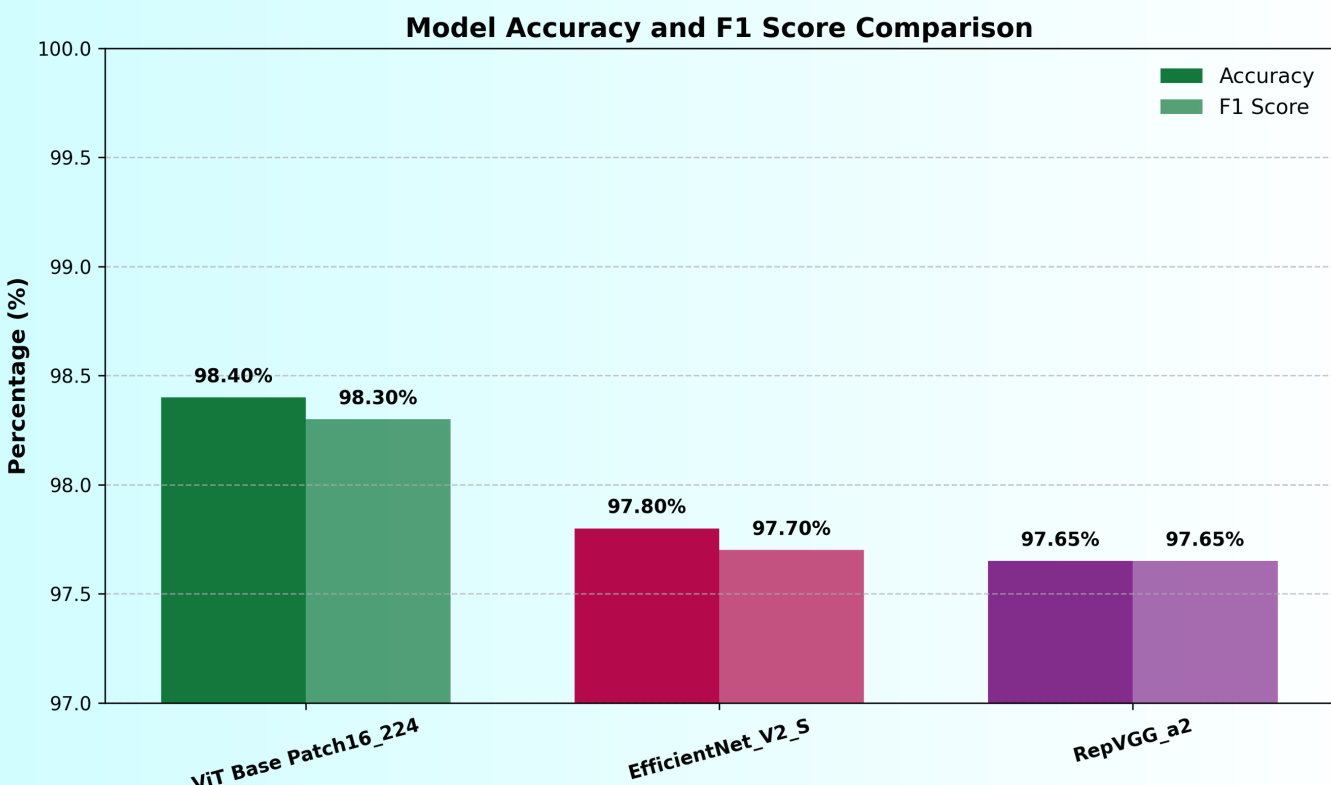
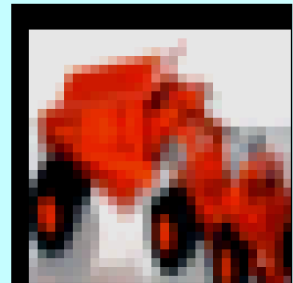
ViT-Base Patch16_224: 86M parameters, CIFAR-10 + ImageNet-21K, 224x224

- ToGray's Structural Precision
- MixUp's Synthetic Generalization
- Dual Color Augmentation (ColorJitter, RandomBrightnessContrast)
- TTA's Ensemble Boost
- CoarseDropout's Spatial Resilience

truck (Original)



truck (Augmented)



04 Fine Tuning

05 Fold-1 Stress Test

06 Hypothesis Challenge

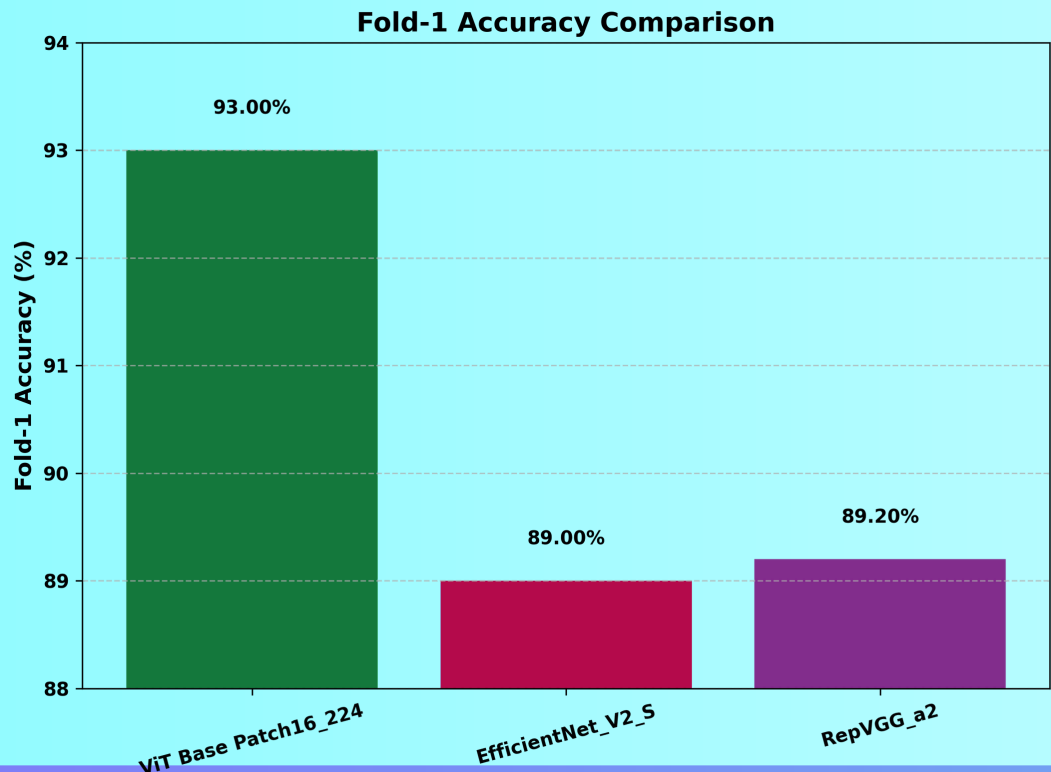
Freezing early layers for 6 epochs preserves low-level features (edges, textures) from pre-training (CIFAR-10/ImageNet)

Unfrozen with $lr=2e-5$, refining mid-level representations (patterns, object parts).

Classifier Regularisation: Sequential (Dropout $p=0.3 \rightarrow$ Linear)

Fold-1 Stress Test

Exposes generalization limits, with ViT's resilience (0.930) a key finding.



The choice of model architecture, combined with strategic augmentations and fine-tuning, drives performance far more than pre-training on CIFAR-10.

E.g., EfficientNetV2-S, without CIFAR-10 pre-training, achieves a strong 97.8% accuracy, slightly edging out RepVGG-A2's 97.65% despite its pre-training.

